

PKCS Standards Summary

	Version	Name	Comments
<u>PKCS #1</u>	2.2	RSA Cryptography Standard ^[1]	See RFC 8017. Defines the mathematical properties and format of RSA public and private keys (ASN.1 -encoded in clear-text), and the basic algorithms and encoding/ padding schemes for performing RSA encryption, decryption, and producing and verifying signatures.
PKCS #2	-	<i>Withdrawn</i>	No longer active as of 2010. Covered RSA encryption of message digests; subsequently merged into PKCS #1.
PKCS #3	1.4	Diffie–Hellman Key Agreement Standard ^[2]	A cryptographic protocol that allows two parties that have no prior knowledge of each other to jointly establish a shared secret key over an insecure communications channel.
PKCS #4	-	<i>Withdrawn</i>	No longer active as of 2010. Covered RSA key syntax; subsequently merged into PKCS #1.
PKCS #5	2.1	Password-based Encryption Standard ^[3]	See RFC 8018 and PBKDF2.
PKCS #6	1.5	Extended-Certificate Syntax Standard ^[4]	Defines extensions to the old v1 X.509 certificate specification. Obsoleted by v3 of the same.
<u>PKCS #7</u>	1.5	Cryptographic Message Syntax Standard ^[5]	See RFC 2315. Used to sign and/or encrypt messages under a PKI . Used also for certificate dissemination (for instance as a response to a PKCS #10 message). Formed the basis for S/MIME , which is as of 2010 based on RFC 5652, an updated Cryptographic Message Syntax Standard (CMS). Often used for single sign-on .

<u>PKCS #8</u>	1.2	Private-Key Information Syntax Standard[6]	See RFC 5958. Used to carry private certificate keypairs (encrypted or unencrypted).
PKCS #9	2	Selected Attribute Types[7]	See RFC 2985. Defines selected attribute types for use in PKCS #6 extended certificates, PKCS #7 digitally signed messages, PKCS #8 private-key information, and PKCS #10 certificate-signing requests.
<u>PKCS #10</u>	1.7	Certification Request Standard[8]	See RFC 2986. Format of messages sent to a certification authority to request certification of a public key. See certificate signing request .
<u>PKCS #11</u>	3	Cryptographic Token Interface[9]	Also known as "Cryptoki". An API defining a generic interface to cryptographic tokens (see also hardware security module). Often used in single sign-on , public-key cryptography and disk encryption ^[10] systems. RSA Security has turned over further development of the PKCS #11 standard to the OASIS PKCS 11 Technical Committee .
<u>PKCS #12</u>	1.1	Personal Information Exchange Syntax Standard[11]	See RFC 7292. Defines a file format commonly used to store private keys with accompanying public key certificates , protected with a password-based symmetric key . PFX is a predecessor to PKCS #12. This container format can contain multiple embedded objects, such as multiple certificates. Usually protected/encrypted with a password. Usable as a format for the Java KeyStore and to establish client authentication certificates in Mozilla Firefox. Usable by Apache Tomcat .
PKCS #13	—	Elliptic-curve cryptography Standard	(Apparently abandoned, only reference is a proposal from 1998.)[12]

PKCS #14	—	Pseudo-random Number Generation	<i>(Apparently abandoned, no documents exist.)</i>
PKCS #15	1.1	Cryptographic Token Information Format Standard ^[13]	Defines a standard allowing users of cryptographic tokens to identify themselves to applications, independent of the application's Cryptoki implementation (PKCS #11) or other API . RSA has relinquished IC-card-related parts of this standard to ISO/IEC 7816-15. ^[14]